Dr. Héctor *Carrión*

World Model Team Lead · Spatio-Temporal & Multimodal AI

Function Health · Ph.D. in Computer Science, UC Santa Cruz / UC Berkeley

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AI research leader with 6+ years of experience building **world models** and self-supervised multimodal representations that capture how physical and biological systems evolve across space, time, and interventions. Currently **World Model Team Lead** at **Function Health**, directing JEPA and autoregressive modeling for longitudinal human health forecasting. Previously **Founding Lead Research Scientist** at **Tera AI** (1B-parameter spatio-temporal world models) and Ph.D. researcher with **Google Research**, **Google X**, and **UC Santa Cruz / UC Berkeley**. First-author work at **MICCAI 2026** (×2), **MICCAI 2023**, **MICCAI MLMI 2022**, and **PLOS Comp. Biol.**, plus granted **Google patents** and open-source models and datasets.

Research interests: world models (JEPA & autoregressive) • spatio-temporal dynamics • self-supervised & multimodal learning • metric 3D & generative priors • longitudinal health forecasting

Research & Professional Experience

Sep 2026 – now
World Model Team Lead, Function Health
- Directing the World Model team and research roadmap on **JEPA** and **autoregressive** multimodal architectures to learn longitudinal representations of member biomarkers, clinical history, and health state.
- Partnering with executive leadership and cross-functional teams to own deliverables from pre-training through trajectory forecasting and personalized health interventions.

2024 – 2026
Founding Lead Research Scientist, Tera AI
- Designed and executed the founding research agenda across generative AI, spatial reasoning, and embodied autonomy, aligning foundation-model experiments with rapid iteration cycles, compute budgets, and product milestones.
- Co-led **TARDIS** and **STRIDE** (with P. Perona and J. Malik): a unified 1B-parameter autoregressive spatio-temporal transformer unifying controllable road-scene synthesis, action prediction, and sequential georeferencing (60% within 10 m).

2020 – 2026
Graduate Student Researcher, UC Santa Cruz & UC Berkeley
- Achieved state-of-the-art performance in label-constrained medical vision (<100 labeled samples) via controllable diffusion (**cgDDI**: 90.9% DDI accuracy, 86.6% EOM; **FEDD**: +18 IoU points) and monocular metric 3D reconstruction (**DermDepth**: correcting depth scale error from 16.1× to 1.11× tuning 0.6% params).
- Built self-supervised temporal stage discovery (**HealNet**) and automated biological imaging pipelines across multi-institutional DARPA/NIH bioelectronic wound-healing studies.

Summer 2023
Student Researcher (Ph.D.), Google Research
- Established foundation-model recipes for unified multimodal video and image pre-training, unlocking >25% higher downstream performance compared to previously deployed single-modality encoders.

2021 – 2022
AI Resident (Ph.D.), X, The Moonshot Factory (Google X)
- Developed simulation, edge-perception, and aerial/ground computer vision systems under novel moonshot problem domains, exceeding target performance benchmarks and earning granted **Google patents**.

Education

2020 – 2026
Ph.D. in Computer Science, University of California, Santa Cruz
Advisor: Dr. Narges Norouzi (UC Berkeley). GPA: 3.92 / 4.00.

2017 – 2021
B.Sc. in Computer Science, University of Puerto Rico, Río Piedras
Undergraduate Research PI: Dr. Rémi Mégret. GPA: 3.75 / 4.00.

2018 – 2019
Software Engineering & Design, Apple Developer Academy, Univ. of Naples Federico II
Full-stack iOS engineering, human-centered interface design, and on-device CoreML applications.

Publications

MICCAI 2026
DermDepth: Toward Monocular Metric Scale 3D Reconstruction Models for Dermatology
H. Carrión, N. Norouzi
16.1× → 1.11× SKINL2 depth scale with 0.6% tuned params paper | arXiv | code | data | model | demo

MICCAI 2026 **cgDDI: Controllable Generation of Diverse Dermatological Imagery for Fair and Efficient Malignancy Classification**
H. Carrión, N. Norouzi
266K+ synthetic images; 90.9% DDI accuracy & 86.6% EOM paper | arXiv | code | data

Comm. Biol. 2025 **A Modular Fluorescent Camera Unit for Wound Imaging**
M. Tebyani, G. Keller, W. S. Hee, P. Baniya, ..., **H. Carrión**, ..., R. R. Isseroff, M. Rolandi, M. Teodorescu
Nature Communications Biology paper

arXiv 2025 **TARDIS STRIDE: A Spatio-Temporal Road Image Dataset and World Model for Autonomy**
★ Co-First
H. Carrión*, Y. Bai*, V. A. Hernández Castro*, K. Panaganti, A. Zenith, M. Trang, T. Zhang, P. Perona, J. Malik
**Equal contribution; 1B-param world model* arXiv | project | code | data

Adv. Mater. Tech. 2025 **Remote-Controlled Wireless Bioelectronics for Fluoxetine Therapy to Promote Wound Healing in a Porcine Model**
H. Li, N. Asefifeyzabadi, ..., **H. Carrión**, N. Norouzi, M. Gomez, M. Zhao, R. R. Isseroff, M. Teodorescu, M. Rolandi
Advanced Materials Technologies paper

Adv. Mater. Tech. 2024 **Programmable Delivery of Fluoxetine via Wearable Bioelectronics for Wound Healing In Vivo**
H. Li, H.-Y. Yang, N. Asefifeyzabadi, ..., **H. Carrión**, ..., N. Norouzi, M. Gomez, M. Zhao, M. Teodorescu, R. R. Isseroff, M. Rolandi
Advanced Materials Technologies paper

MICCAI 2023 **FEDD: Fair, Efficient, and Diverse Diffusion-Based Lesion Segmentation and Malignancy Classification**
H. Carrión, N. Norouzi
81% DDI malignancy accuracy from 60 labeled images paper | arXiv | code

MICCAI-W 2022 **HealNet: Self-Supervised Acute Wound Heal-Stage Classification**
H. Carrión, M. Jafari, H.-Y. Yang, R. R. Isseroff, M. Rolandi, M. Gomez, N. Norouzi
Machine Learning in Medical Imaging (MLMI); 90.6% stage accuracy paper | arXiv

PLOS CB 2022 **Automatic Wound Detection and Size Estimation Using Deep Learning Algorithms**
H. Carrión, M. Jafari, M. D. Bagood, H.-Y. Yang, R. R. Isseroff, M. Gomez
PLOS Computational Biology paper | code

VISAPP 2022 **Honeybee Re-Identification in Video: New Datasets and Impact of Self-Supervision**
J. Chan, **H. Carrión**, R. Mégret, J. Rivera, T. Giray
109K-image self-supervised track dataset paper

Open-Source Code, Models & Datasets

DermDepth	Metric 3D weights, D-Synth dataset, and live Space demo	github.com/hectorcarrion/dermdepth
cgDDI	Hybrid diffusion code and 266K+ synthetic dermatology dataset	github.com/hectorcarrion/ControllableGenDDI
TARDIS	1B-param world model (JAX/EasyLM) and STRIDE dataset	github.com/tera-ai/tardis
FEDD	Diffusion feature probes and DDI lesion segmentation masks	github.com/hectorcarrion/fedd

Honors, Patents & Advisory

Patents	**Google Patent Award** — Granted perception & autonomous systems patents at X (Google X)
Fellowships	**Google–CAHSI Doctoral Fellowship** | **NIH BD2K Biomedical Data Science Fellowship**
Awards	**MICCAI STAR Award** | **SIDIM Outstanding Research Award**
Advisory	**Technical Advisor**, Visia AI (2021–present) — computer vision for recyclability & hazard intelligence
Mentorship	**Machine Learning Tutor**, UC Santa Cruz (2020–2022) — mentored students in deep learning & computer vision

Skills & Languages

Programming	Python, C++, Swift, JAX, Bash, LaTeX
Deep learning	PyTorch, JAX/EasyLM, TensorFlow, Keras, JEPA & autoregressive world models, diffusion, multimodal LLMs
Domains	Longitudinal health forecasting, metric 3D reconstruction, medical imaging, spatial autonomy, fairness
Leadership	Research strategy & team management, stakeholder alignment, multi-node cluster budgeting, product design
Languages	English (native), Spanish (native)